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Purification of molt-inhibiting hormone-like peptides with hyperglycemic activity from the eyestalks of the crab *Scylla serrata*

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ABSTRACT: In the present study, putative molt-inhibiting hormone (MIH)-like peptides from the sinus glands of *Scylla serrata* (*S. serrata*) were characterized by high-performance liquid chromatography, followed by immunoassay and bioassay (inhibition of ecdysteroid synthesis by Y-organs). Results of double immunodiffusion demonstrated the cross-reactivity of *Homarus americanus* MIH antiserum with the *S. serrata* MIH-like peptides, indicating the existence of close immunological relationship between crab and lobster peptides. Of the three molt-inhibiting immuno-reactive peptides, two also possess significant hyperglycemic activity.

KEY WORDS: **ecdysteroid, eyestalks, hemolymph sugar, molt-inhibiting hormone, *Scylla serrata*.**

INTRODUCTION

In order to increase in size, crustaceans must first shed (molt) their confining exoskeleton. This molting process is regulated by ecdysterioids, secreted from the molting gland (Y-organ).[1] The synthesis and secretion of ecdysteroids by the Y-organ appears to be under the inhibitory control of the neuropeptide molt-inhibiting hormone (MIH) secreted by the X-organ-sinus gland complex within eyestalks.[2–4] The MIH acts by inhibiting steroidogensis through high-affinity specific receptors on Y-organ cell-membrane.[5]

Although hormonal regulation of molting has been intensively investigated in several decapod crustaceans,[6–13] only few reports are available on the purification and characterization of hormones from the commercially important species. Though the presence of MIH in crustacean eyestalks has been known for more than 90 years, the hormone has been fully characterized from sinus glands of only a few species. A putative MIH was isolated from the shore crab *Carcinus maenas*,[5,14] the American lobster *Homarus americanus* (*H. americanus*),[15] the kuruma prawn *Penaeus japonicus*,[16] the sand prawn *Metapenaeus ensis*[17] and recently from the crayfishes *Procambarus clarkii* (*P. clarkii*)[18,19] and *Procambarus bouvieri* (*P. bouvieri*).[20] The cDNA for MIH was also isolated from the sinus glands of several crusatacean species.[21–24] The MIH peptide of *P. clarkii* was also synthesized chemically.[25,26] Chang *et al.*[15] also described a bioassay for MIH, and reported the complete amino acid sequence of *Homarus* MIH. This paper describes the isolation and characterization of MIH-like peptides from the sinus glands of *Scylla serrata* (*S. serrata*). An unique feature of these peptides is that two of the three peptides have both molt-inhibiting and hyperglycemic functions.

MATERIALS AND METHODS

Animals

Scylla serrata (50–60 g body weight) were collected from the Pondicherry coast and kept in the laboratory recirculated sea water (salinity 33 ppt; temperature 27–29°C) under a natural photo period. Molt stages were determined by microscopic examination of setogenesis in the exopodite of third maxilliped.

Preparation of sinus gland extract

Sinus glands were dissected from freshly excised eyestalks from 100 intermolt (Stage C_4) crabs. Homogenates were prepared by grinding the 200

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sinus glands in 0.1 N HCl (400 μL), with a glass-glass micro homogenizer. After heating for 5 min at 80°C, the extracts were centrifuged at 16 000 × *g* for 5 min.

The sinus gland extract was further resolved on high performance liquid chromatography (HPLC) using C_{18} μBondapac column (Waters, Milford, MA, USA) with a linear gradient of 20–80% methanol in 0.1% triflouroacetic acid over a period of 2 h at a flow rate of 1.0 mL/min. Peptide peaks were detected by their absorbance at 280 nm. Fractions were collected and used for bioassay.

Immuno-cross reactivity determination

The double diffusion experiments were performed according to the published procedure of Ouchterlony[27] using 0.8% agarose in a 10 mM sodium phosphate buffer, pH 7.0. In brief, aliquots of fractions (four sinus gland equivalents) were analyzed for precipitin formation using antiserum raised in rabbits against *H. americanus* MIH (1 : 100 dilution). Earlier, Nagaraju[28] used *H. americanus* MIH antibodies to detect MIH-like peptides from the sinus glands of the freshwater crab *Oziotelphusa senex senex*. The *H. americanus* MIH antiserum was generously supplied by Professor E. S. Chang (University of California at Davis, USA). Fractions that cross-reacted with antiserum were further tested in an *in vitro* Y-organ assay and an *in vivo* bioassay for hyperglycemic activity.

In vitro Y-organ (ecdysteroid) secretion

Y-organs were isolated from *S. serrata*. Each Y-organ was cut into two pieces and preincubated in sterile crab saline[29] (containing 30.68 g sodium chloride, 0.989 g potassium chloride, 1.375 g calcium chloride, 2.359 g magnesium chloride and 0.222 g sodium bicarbonate in 1 L of double distilled water; pH 7.0) for 30 min. Since Y-organ pieces did not survive for more than 6 h in crab saline,[29] modified medium 199 was used to culture the Y-organ pieces. Though medium 199 was devised for vertebrate tissues, Y-organ pieces cultured in modified medium 199 did not show any symptoms of cell toxicity/injury. The incubated Y-organ cells were observed under a microscope and no significant changes were detected in the size and shape of the cells. The cells were normal even after 72 h incubation in the culture medium. Individual Y-organ pieces were incubated in 24-well tissue culture plates in 500 μL of modified medium 199 containing fetal bovine serum (10%), penicillin-G (12.5 μg/mL), streptomycin (12.5 μg/mL) and either 4 μL of peptide fraction (one sinus gland equivalent peptide) or 4 μL of sinus gland extract containing 1.0, 2.0, or 5.0 sinus gland equivalents. Modified medium 199 was prepared by dissolving 10.5 g of dried 199 medium (containing Earle's salts, L-glutamine without sodium bicarbonate) in 1 L of distilled water, and sodium bicarbonate (2.2 g) was added to the medium after sterilization to obtain pH 7.0. Controls were maintained with 4 μL of non-sinus gland (2% chelate leg muscle tissue) extract instead of peptide fraction. After 12 and 24 h incubation in CO_2 incubator at 20°C, 50 μL of the medium was removed from each well and extracted in 75% methanol and centrifuged (4000 × *g*, 10 min). The supernatant was removed and the pellet was washed again with 75% methanol and centrifuged. The methanolic supernatants were combined and stored at –20°C. The radio immunoassay was conducted as previously described[30,31] using the antiserum generously supplied by Dr W. E. Bollenbacher (University of North Carolina, Chapel Hill, USA). Percent inhibition of ecdysteroid secretion was calculated as described elsewhere.[3]

Crustacean hyperglycemic hormone bioassay

Bilaterally eyestalk ablated (3 days) intact male crabs (body weight 30–35 g) were used for crustacean hyperglycemic hormone (CHH) bioassay. Each crab was injected with 10 μL of either purified peptide or sinus gland extract (containing 1.0, 2.0, or 5.0 sinus gland equivalents). Control crabs received 10 μL of non-sinus gland (2% chelate leg muscle) extract. Injections were made through the arthrodial membrane at the base of the coxa of the third pair of walking legs. Hemolymph (50 μL) was taken just prior to injection from the base of the fourth pair of walking leg. After 120 min, a second hemolymph sample (50 μL) was collected and the glucose concentration in the hemolymph was estimated using glucose enzyme reagent (Sigma, Missouri, USA).

Data analysis

The data were analyzed statistically using Dunnett's test to determine differences between control and treated groups.

RESULTS

A typical chromatogram from HPLC separation of crab sinus gland extracts is shown in Figure 1. The

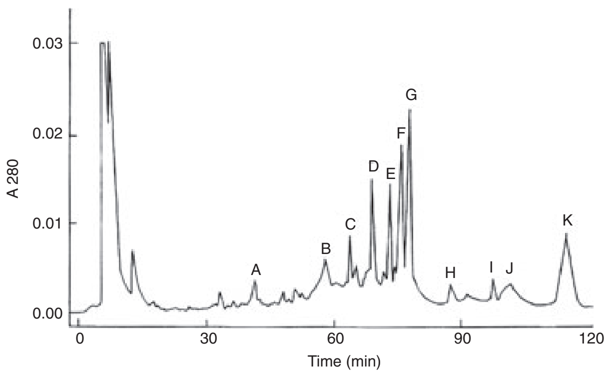

Fig. 1 High performance liquid chromatogram of *Scylla serrata* sinus gland peptides. Solvent A, 0.1% trifluoroacetic acid in water; solvent B, methanol containing 0.1% trifluoroacetic acid. A linear gradient of 20–80% solvent B for 120 min was used. Flow rate was 1.0 mL/min. Peptide peaks were detected by their absorbance at 280 nm. Peaks D, F and G were shown to react with *Homarus* molt-inhibiting hormone antiserum.

Table 1 Immunological cross reactivity of *Scylla serrata* eyestalk peptides with antisera raised against *Homarus* molt-inhibiting hormone

Peptide peaks	Immuno-cross reactivity
A	–
B	–
C	–
D	++++
E	–
F	+++
G	+++
H	–
I	–
J	–
K	–

+, Immuno-cross reactivity.

chromatogram shows the presence of a number of peaks with the majority eluting after 60 min. Each protein peak eluting was collected and used for their ability to cross react with *Homarus* MIH antibodies.

The immunological cross reactivity of the different peptides eluted towards *Homarus* molt-inhibiting peptide antibodies is shown in Table 1. Among all the fractions (A-K), only three peaks (D, F, G) showed immuno-cross reactivity towards *Homarus* antibodies that were eluted between 60 and 90 min. The major secretary product of *Scylla* Y-organ (in *in vitro* studies) was ecdysone, though traces of 3-dehydroecdysone was also observed (unpubl. data, P S Reddy & P R Reddy 2005). The effect of each immunoreactive fraction on ecdysteroid secretion from Y-organ *in vitro* was determined after 12 and 24 h, and is shown in Table 2. It is clear from the data, that peaks F and G are very potent in inhibiting Y-organ to secrete ecdysteroid. The longer incubation time appeared to be more potent in inhibiting ecdysteroid secretion from Y-organ. The antibodies used in the present study did not react with the 3-dehydroecdysone.

The three peptide fractions were also tested for hyperglycemic activity in eyestalk-ablated crabs and the results are presented in Table 3. Among the three peptides, peaks F and G contained significant hyperglycemic activity, although with varying degrees of potency. The large peak (peak G) contained most hyperglycemic and also contained significant molt-inhibiting activity.

Table 2 Ecdysteroid secretion by Y-organs of *Scylla serrata* in the presence of non-sinus gland extract, sinus gland extract and sinus gland peptide fractions

	Ecdysteroid secretion (percent of control)	
Sample	12 h	24 h
NSG	100.0 ± 8.5	98.6 ± 4.7
SGE		
1.0	62.9* ± 28.0	52.9* ± 9.0
2.0	55.4* ± 9.78	57.7* ± 7.31
5.0	50.3* ± 8.65	55.4* ± 6.57
D		
1.0	83.6† ± 21.0	62.4* ± 5.8
2.0	77.4* ± 12.3	79.3** ± 7.65
5.0	61.7* ± 11.4	68.1* ± 8.77
F		
1.0	38.2* ± 12.0	46.6* ± 9.8
2.0	24.5* ± 8.78	27.3* ± 7.65
5.0	20.1* ± 9.83	24.5* ± 8.32
G		
1.0	48.5* ± 8.2	56.1* ± 10.0
2.0	47.9* ± 7.68	54.9* ± 9.92
5.0	41.4* ± 9.83	52.5* ± 10.7
F-value	14.625	26.743
P	<0.0001	<0.0001

Y-organ pieces were incubated with either sinus gland extract or peptide fraction for 12 and 24 h at 20°C. Each value represents mean ± standard deviation of five samples. $^{*}P < 0.05$; $^{**}P < 0.001$; †not significant when compared with non-sinus gland (NSG) extract. SGE, sinus gland extract.

Table 3 Hyperglycemic activity of *Scylla serrata* sinus gland extract and sinus gland peptides assayed in eyestalk ablated male crabs

Sample	Glucose concentration (mg/mL)	
	Before injection	120 min after injection
Control	0.32 ± 0.07	0.34 ± 0.06 (6.25)
SGE		
1.0	0.35 ± 0.08	0.48* ± 0.08 (37.14)
2.0	0.34 ± 0.08	0.53* ± 0.17 (55.88)
5.0	0.34 ± 0.09	0.70* ± 0.12 (105.88)
D		
1.0	0.36 ± 0.11	0.38 ± 0.08 (5.55)
2.0	0.35 ± 0.12	0.38 ± 0.18 (8.57)
5.0	0.34 ± 0.09	0.35 ± 0.09 (2.94)
F		
1.0	0.35 ± 0.14	0.53* ± 0.17 (51.42)
2.0	0.34 ± 0.11	0.77* ± 0.21 (126.47)
5.0	0.37 ± 0.10	0.96* ± 0.23 (159.45)
G		
1.0	0.32 ± 0.08	0.95* ± 0.32 (196.87)
2.0	0.33 ± 0.08	1.04* ± 0.66 (215.15)
5.0	0.35 ± 0.07	1.22* ± 0.14 (248.57)
F-value	0.1792	11.80
P	= 0.9989	<0.0001

The hemolymph glucose level was measured both immediately before and 120 min after injection of the sample (sinus gland equivalents). Control crabs were injected with 10 μL of non-sinus gland extract. Values are mean ± standard deviation of eight crabs. *Values for sinus gland extract and peaks F and G were significantly different from controls ($P < 0.001$). Values in parenthesis are percentage change from control.

DISCUSSION

The identification of MIH-like peptides in *S. serrata* sinus gland extract was made easier by the use of an antiserum generated against *Homarus* MIH. The cross-reactions of *Scylla* peptides with *Homarus* MIH antibodies indicate a close immunological relationship of peptides between lobsters and crabs.

All the peptides that cross-reacted with *Homarus* antibodies contained molt-inhibiting bio-activity with differential potency. Although all the three peptides showed some molt inhibiting bio-activity in the *in vitro* Y-organ assay, only two of these contained significant hyperglycemic activity. Chang *et al.*[32] purified, from eyestalks of *H. americanus*, a peptide that has both molt-inhibiting and hyperglycemic activity. This peptide appears to be identical (96% amino acid sequence homology) to one of the isoforms of hyperglycemic hormone from the same species, and it has therefore been reasonably suggested that one of the hyperglycemic hormone isoforms act as MIH. Due to the high degree of similarity of the amino acid sequence of MIH and CHH, it is not surprising that MIH has hyperglycemic activity.

The data is also in agreement with the presence of different isomorphs of CHH found in crayfishes *P. bouvieri*,[33] *Orconectes limosus* (*O. limosus*)[34] and *Astacus leptodactylus* (*A. leptodactylus*),[35] and in lobster *H. americanus*.[36] In *P. bouvieri*,[33] *O. limosus*[34] and *H. americanus*,[36] two isomorphs of hyperglycemic hormone were determined, but more than two isomorphs of hyperglycemic hormone were identified in *A. leptodactylus*.[35] Yang *et al.*[37,38] identified five molecular species of hyperglycemic hormones from a single prawn *Penaeus japonicus* and suggested gene multiplication regarding the hyperglycemic hormone family. Huberman *et al.*[39] discovered two isomorphs of hyperglycemic and MIH in *P. bouvieri* that only differ from parental hormones by post-translational modification of the third residue for L-Phe to D-Phe. Ohira *et al.*[40] isolated six hyperglycemic hormones and one MIH from the sinus gland of kuruma prawn *Penaeus japonicus*. In *Penaeus japonicus*, Shih *et al.*[41] demonstrated that not all of hyperglycemic hormone family peptides were produced in one neurosecretory cell cluster, and suggested different physiological regulatory factors for hyperglycemic family peptides.

The results provide the first evidence that the presence of three immunoreactive peptides with molt-inhibiting activity in the crab *S. serrata*. Recently, while characterizing hyperglycemic hormone from *P. clarkii*, Yasuda *et al.*[42] demonstrated that one of the two variants of hyperglycemic hormone exhibited more potent molt-inhibiting activity than hyperglycemic activity. These results have not only suggested the unsuspected complexity in the endocrine control of crustacean molting, but also leads to the question of whether the correct primary functions of these peptides have been identified. It is not clear if the inhibition of steroidogenesis in Y-organs is the primary function of MIH. It is also clear that methyl farnesoate (MF), a secretary product from the mandibular organ, stimulates the Y-organ to secrete ecdysterioids.[43] The secretion of MF is under the negative control of eyestalk peptide hormones.[44–48] Eyestalk ablated lobsters *H. americanus* display increased levels of both MF and ecdysteroids.[1] This would indicate that at least two structurally diverse hormones (peptide and terpenoid) are involved in the negative and positive regulation of ecdysteroid secretion by the Y-organ.

The dual biological functions in the peptide may be due to a variation in an amino-acid side-chain and is consistent with the observations in the

Mexican crayfish.[20] In the present study, designation of these peptides as MIH and CHH is a consequence of the observed differences in activity. Furthermore, physiologically-based studies are now required to define more clearly the role of hyperglycemic hormones in molt regulation. Because some of these peptides have at least dual biological activities, the designation of any one of these peptides as the MIH or the CHH is currently premature.

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